

.

EE1060: Differential Equations and Transform Techniques

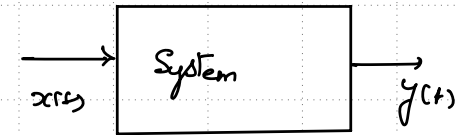
Lecture 28: Convolution

Topics :

1. Time Invariance
 2. Convolution
-

Time Invariance

A system is Time Invariant if the response to

$$x(t-\tau) \text{ is } y(t-\tau)$$


Example: a) $y(t) = 2x(t) + 4$ $\left. \begin{array}{l} y(t) = 2x(t-\tau) + 4 \\ y(t-\tau) = 2x(t-\tau) + 4 \end{array} \right\} \text{ TI \& not linear.}$

b) $y(t) = tx(t)$ $\left. \begin{array}{l} y_1(t) = tx_1(t) \\ y_2(t) = tx_2(t) \end{array} \right\} \begin{array}{l} x_1 + x_2 \Rightarrow y = t(x_1 + x_2) \\ = y_1 + y_2 \end{array}$
 linear & not TI.

$\left. \begin{array}{l} y(t) = tx(t-\tau) \\ y(t-\tau) = (t-\tau)x(t-\tau) \end{array} \right\} y(t) \neq y(t-\tau)$

Any system whose i/p-o/p relation is governed by an ODE (with constant coefficients) is Time Invariant.

Time Invariance

Any system whose i/p-o/p relation is governed by an ^{nth order} ODE (with constant coefficients) is time invariant.

$$a_n \frac{d^n y}{dt^n} + \dots + a_1 \frac{dy}{dt} + a_0 y = x(t) \rightarrow \textcircled{1} \quad y(t) \text{ sol of } \textcircled{1}$$

$\underbrace{\hspace{10em}}_{L[y] = x(t)}$

$y(t-\tau)$ is a sol to $L[y] = x(\underline{t-\tau})$

replacing t with $t-\tau \Rightarrow y(t-\tau)$ is a sol to $L[y] = x(t-\tau)$
 rep $t_1 \rightarrow t$

Convolution for LTI Systems

Response of a Linear System

- ↳ Breaks the i/p into **Sampler signals**
- ↳ resp as linear Comb of the response of the system to the sampler signals.

Synthesis Eqn:-

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

a) $h_{\tau}(t) \rightarrow$ impulse response of system to the i/p $\delta(t-\tau)$

Overall resp $\rightarrow$ resp to pulse signals (Σ pulse signals $\rightarrow$ i/p signal)

a) $h_{K\Delta}(t) \rightarrow$ response to the pulse $\delta_p(t-K\Delta)$ | unit pulse $\rightarrow \delta_p(t-K\Delta)\Delta$

b) Scale of width $x(K\Delta) \rightarrow x(K\Delta) h_{K\Delta}(t) \Delta$

c) Sum due to the resp $\Rightarrow y_p(t) = \sum_{K=-\infty}^{\infty} x(K\Delta) h_{K\Delta}(t) \Delta$

$$\lim_{\Delta \rightarrow 0} y_p(t) = y(t) = \int_{-\infty}^{\infty} x(\tau) h_{\tau}(t) d\tau$$

Example and Convolution:-

$h_c(t) \rightarrow$ impulse resp to $\delta(t-\tau)$

If Sys is TI:- $h_c(t) = h(t-\tau)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \Rightarrow \text{Convolution } x(t) * h(t)$$

What is wrong with this solution } :- $\frac{dy}{dt} + ay = x(t) \rightarrow h(t) = e^{-at}$

$$x(t) = \sin \omega t \Rightarrow y(t) = \int_{-\infty}^{\infty} \sin(\omega \tau) e^{-a(t-\tau)} d\tau.$$

$$= e^{-at} \int_{-\infty}^{\infty} \sin(\omega \tau) e^{a\tau} d\tau \rightarrow \text{does this limit converge?}$$

$$x(t) = \sin \omega t \cdot u(t) \rightarrow y(t) = e^{-at} \int_0^{\infty} \sin(\omega \tau) e^{a\tau} d\tau \rightarrow \text{does this limit converge?}$$